

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

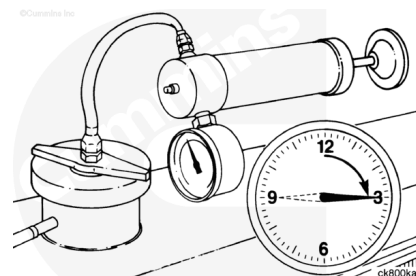
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-035.html)

- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-089.html>)

Leak Test

Pressurize the engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)

Note : Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Be sure the system is holding air pressure before beginning the inspection.



Measurements

	kpa	psi
Air Pressure:	138	20

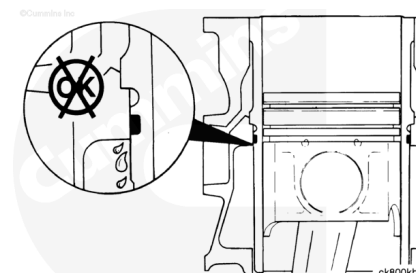
Inspect the outside circumference of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a coolant leak is found:

- Remove and inspect the cylinder liners. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-028.html>)

Refer to Service Bulletin, Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753 (</qs3/pubsys2/xml/en/bulletin/4383753.html>), to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

- Inspect the cylinder block liner bore area. Refer to Procedure 001-058 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-058.html>)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-089.html>)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-035.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)
- Fill and prime the engine lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks.

If a malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.

- Use the following procedure to check the aftertreatment selective catalytic reduction. Refer to Procedure 014-025 in Section 14. (</qs3/pubsys2/xml/en/procedures/377/377-014-025.html>)
- Use the following procedure to check the aftertreatment diesel particulate filter. Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)